Craftax runs compared: achievements and what unlocks them

The taste family on family/taste, with the plain baseline and the no-taste control of next-taste: 1.1M env steps each, size50m, scored on the same 30 fixed worlds with tools/death_eval.py. (M) = masked: the run uses the rule mask, which reads the game's action rules, so it counts as cheating. Updated 2026-11-08.

Every number is the mean episode-final achievement count over 30 evaluation episodes, ± its standard error over those worlds; unlock rates are the share of episodes in which an achievement fired; return is the mean raw Craftax return as a % of the maximum (226). Every checkpoint is scored on the same fixed worlds (env.reseed, audit B05), so comparisons are paired world by world. The error bars cover the worlds only, not training seeds.

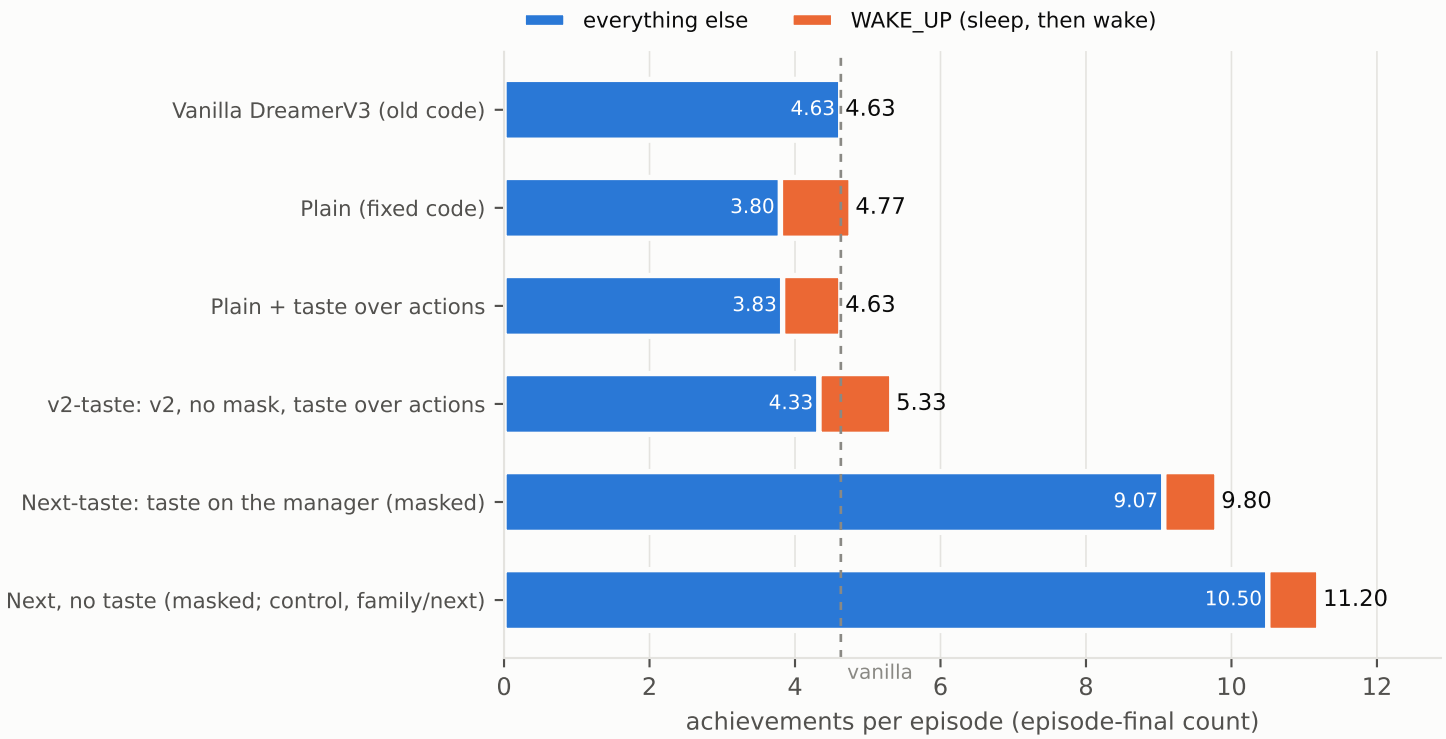
Taste did not help anywhere it was tried. Over actions on plain DreamerV3 (abl_taste) it scores 4.63 against plain's 4.77 (-0.13 ± 0.27, paired). In place of v2's rule mask (v2_taste) it scores 5.33, above vanilla (+0.70 ± 0.28) and level with its no-taste control in training. On the manager of the next agent (next_taste, masked) it scores 9.80, against 11.20 for the same agent with taste switched off (next_lam0, masked); over 60 worlds 9.38 against 11.30, -1.92 ± 0.41.

run	ach. ± SE	w/o wake	life	return	table	pickaxe	sword	stone	drink
Vanilla DreamerV3 (old code)	4.63 ± 0.22	4.63	295	1.65%	63%	23%	3%	0%	77%
Plain (fixed code)	4.77 ± 0.28	3.80	265	1.71%	50%	3%	10%	0%	73%
Plain + taste over actions	4.63 ± 0.29	3.83	262	1.65%	50%	0%	13%	0%	43%
v2-taste: v2, no mask, taste over ac	5.33 ± 0.19	4.33	280	1.96%	70%	0%	0%	0%	87%
Next-taste: taste on the manager (r	9.80 ± 0.42	9.07	287	3.94%	100%	90%	60%	73%	93%
Next, no taste (masked; control, far	11.20 ± 0.41	10.50	284	4.56%	100%	100%	83%	97%	77%

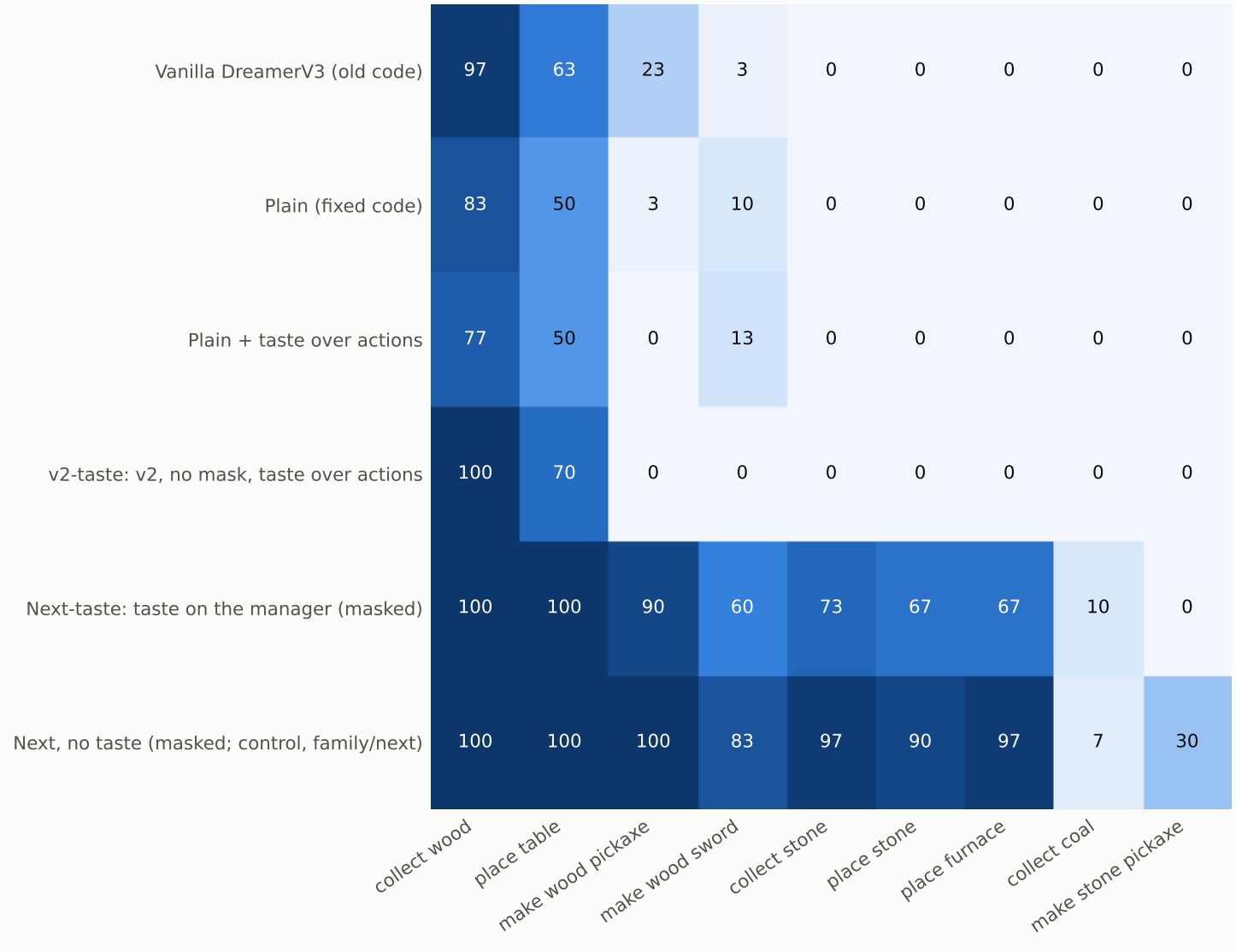
ach. ± SE = mean episode-final achievements ± standard error over the 30 worlds (one training seed). w/o wake = the same minus WAKE_UP. life = mean lifespan in steps. return = raw Craftax return, % of 226. Rates are % of episodes. Vanilla: the original checkpoint (code before the 2026-10 fixes), re-scored on the fixed worlds. Plain: main's code, --configs craftax size50m --env.craftax.survival none (seed 0). The taste runs were trained on the fixed code (abl_taste and next_taste on 49a6bc3, v2_taste on cbec67b); their exact flags are in README.md. (M) = masked: uses the rule mask, counted as cheating, so not comparable with the rest. Next, no taste is the no-taste control of next-taste (taste computed but lam_mgr 0); the run lives on family/next.

Achievements per run, and the early tech tree

Mean episode-final achievements per run, split into WAKE_UP and the rest; unlock rate of the early tech-tree rungs

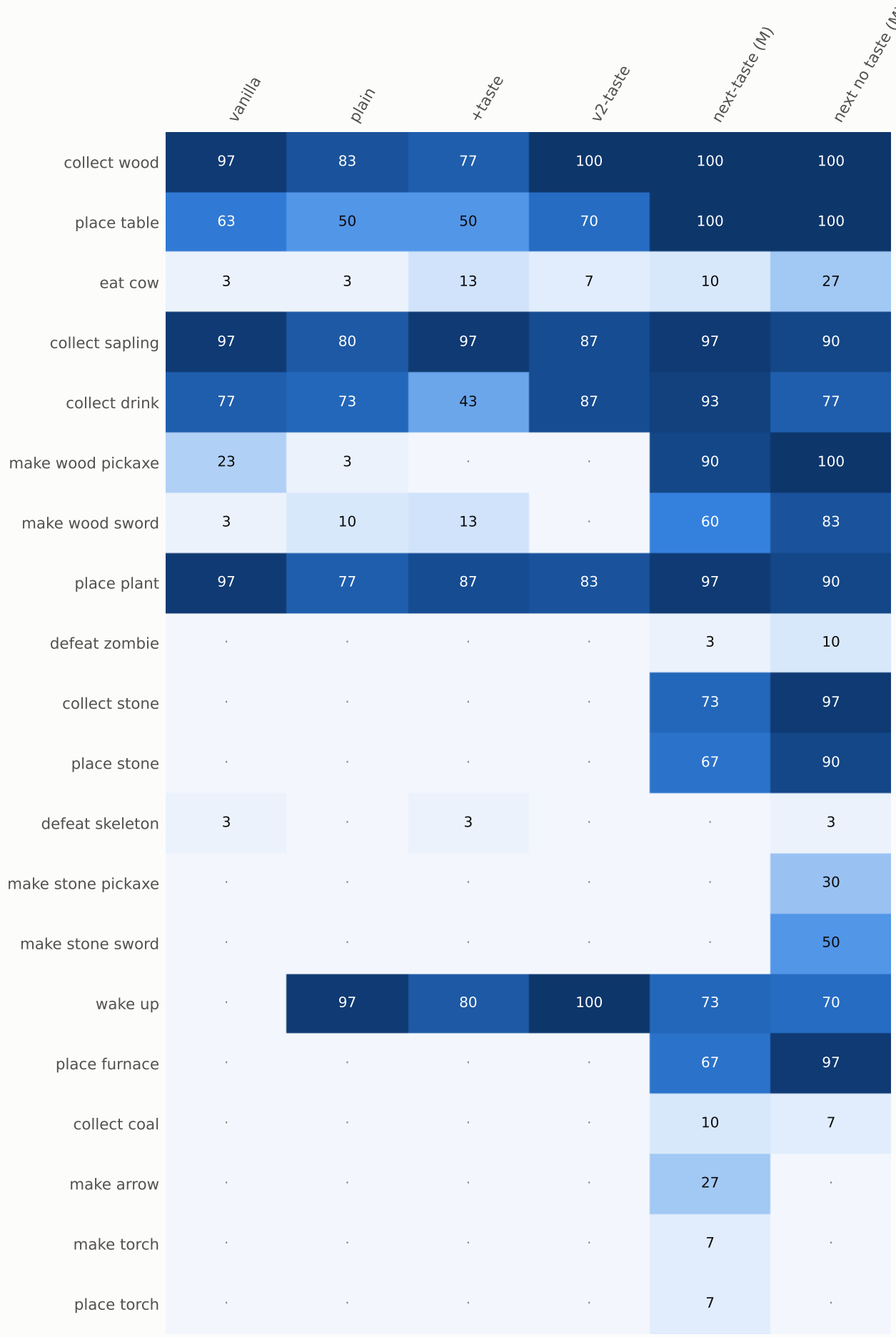


Unlock rate (% of episodes), early tech tree in dependency order



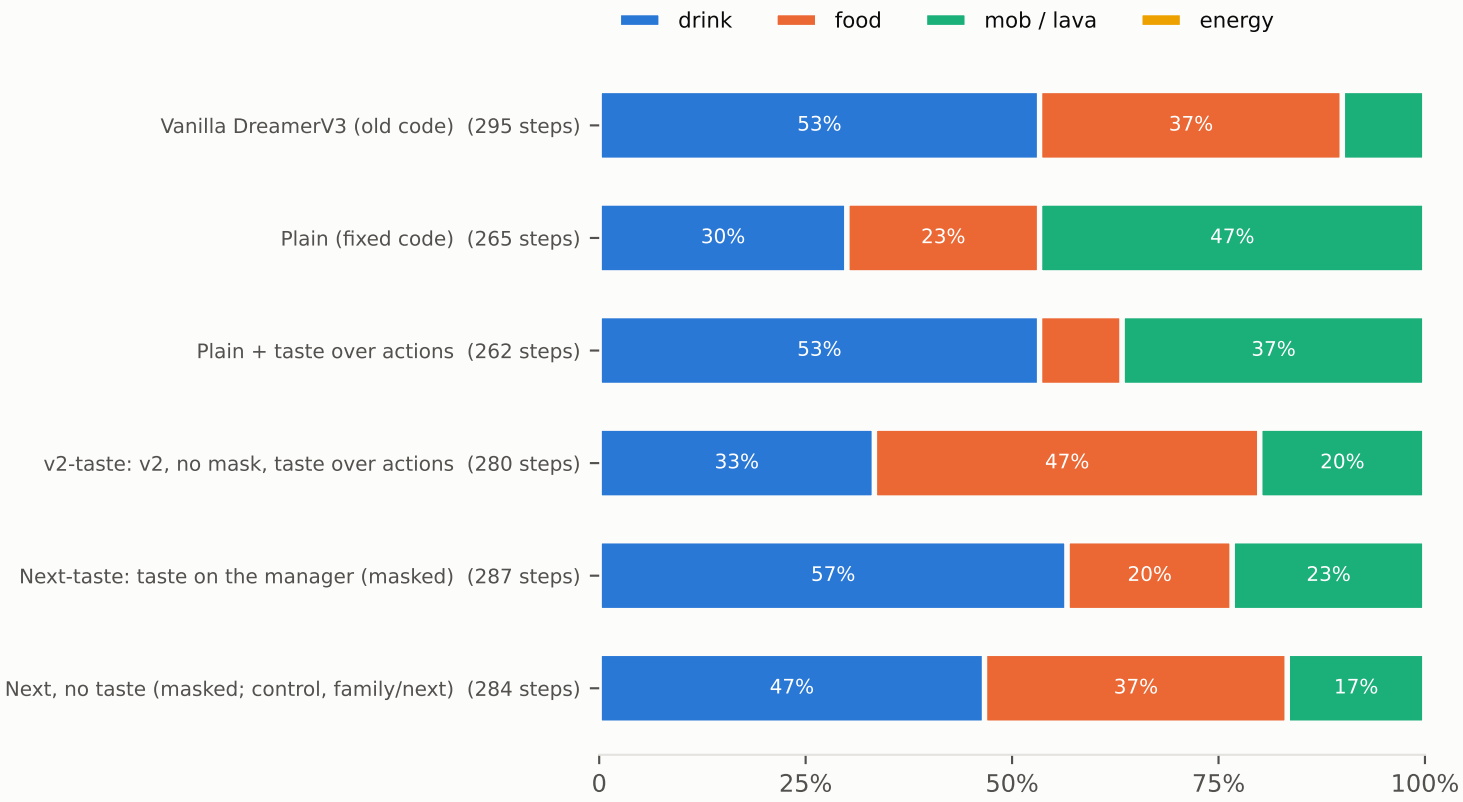
Every achievement any run unlocked

% of evaluation episodes in which each achievement fired. The other 47 of 67 were never unlocked by any run.



Cause of death and lifespan

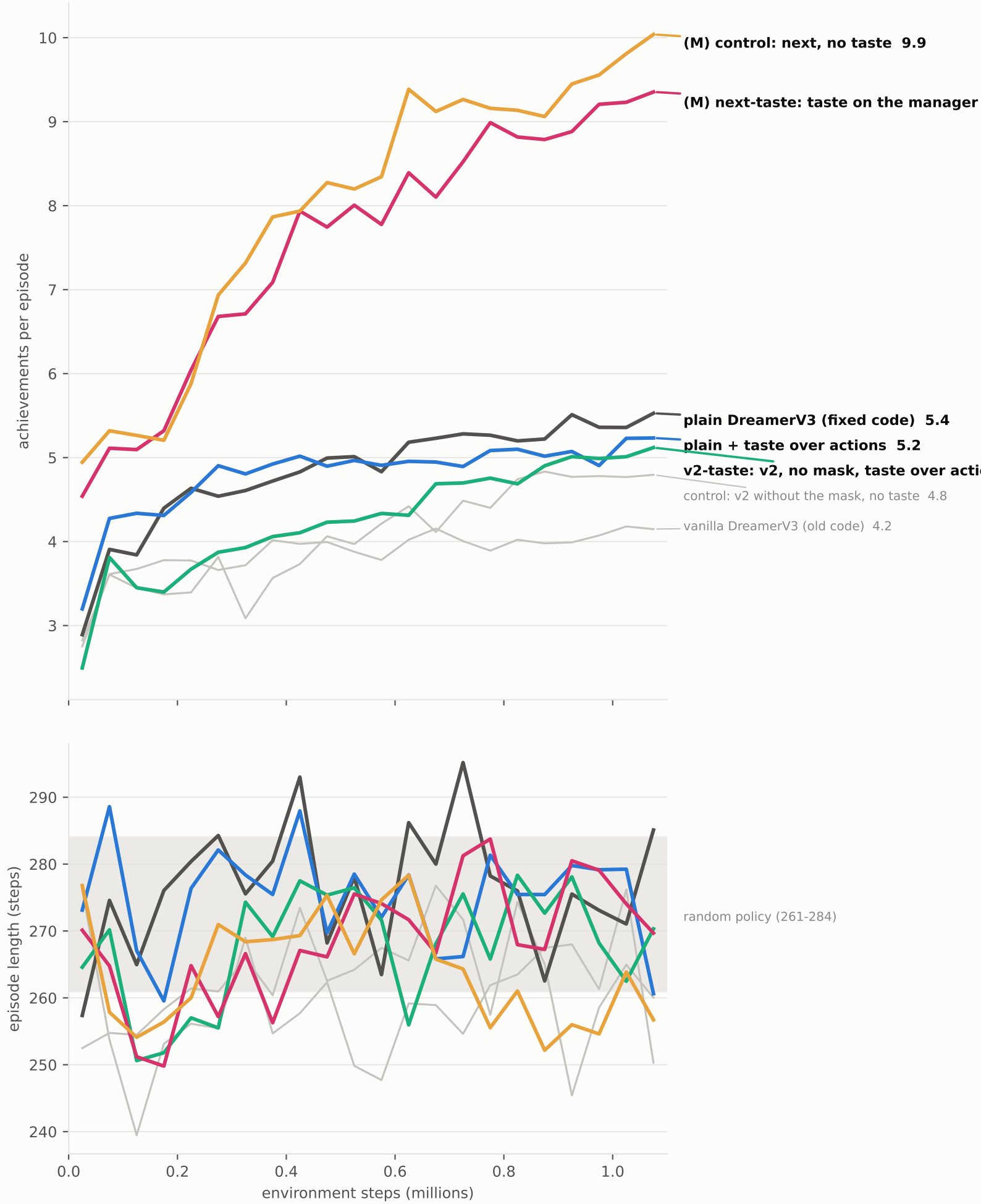
Cause of death (share of evaluation episodes) and lifespan; and, if the results list one, how much a run moves between its own checkpoints



Random policy lives 261-284 steps; doing nothing lives 333.

Training curves: every run

Rebuilt from each run's replay; 50k-step bins; value = mean of the last 100k steps.



What the results say, and what would settle the rest

Taste over actions: no effect, with or without the rest of the agent

On plain DreamerV3, centered taste over actions scores 4.63 against 4.77 (-0.13 ± 0.27 paired by world) and trains slightly lower (5.19 against 5.43 over the last 100k steps). In place of v2's rule mask (v2-taste, uncentered evidence) it scores 5.33: +0.70 ± 0.28 over vanilla, but in training level with v2 without the mask and without taste (v2-norules, not scored). Uncentered, its bias came to follow how often a button was pressed (rank correlation -0.98): placing a table, which v2 needs, ended as disliked as an iron sword it cannot make.

Taste on the manager cost the next agent about two achievements (masked runs)

Next-taste locked the manager onto one goal, place table, at its first update (6.7k steps) and held it to the end: only a chosen goal gets evidence, centering over the chosen goals leaves the winner's evidence near zero, and taste outweighed the critic's spread between goals 15 to 300 times over. Against the same agent with taste switched off it is -1.92 ± 0.41 over 60 worlds; it never made a stone tool, while the control made the stone sword in half its episodes on worlds 0-29. Both runs use the rule mask, so neither is comparable with knowledge-free agents. docs/taste-prior.md, "Why taste hurt".

What would settle the rest

Every taste result is one seed. If taste is kept: put it on the critic's scale (units of the critic's own spread between goals), let evidence reach goals the manager did not choose or decay back to neutral, treat "no goal" like the other goals, and test it without the rule mask.